



**BALL STATE  
UNIVERSITY**

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# **Module 9 Overview & Checklist**

Introduction to Statistical Methods

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# 1 Learning Objective(s)

## 1.1 Textbook Learning Objectives

- Differentiate between Type I and Type II Errors
- Describe hypothesis testing in general and in practice
- Conduct and interpret hypothesis tests for a single population mean, population standard deviation known.
- Conduct and interpret hypothesis tests for a single population mean, population standard deviation unknown.
- Conduct and interpret hypothesis tests for a single population proportion.

## 1.2 Instructor Learning Objectives

- Be able to identify different parts of a hypothesis given a narrative example
- Understand the connection between  $\alpha$ , p-values, probability, and Type I and II errors
- Be able to work through practical inferential statistics computations for a single sample scenario

# 2 Assessments

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Quiz 8 on Module 8 Content on Canvas

# 3 Lecture(s) & Participation

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Watch Module 9 Lecture on Canvas and Take Notes

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Complete and Turn-in Module 9 Lecture Check-in on Canvas

# 4 Reading(s)

*Under "Readings" in Canvas Module*

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Read Chapter 9 of Textbook

## 5 Homework & Assignment(s) Due

*Under "Assignments" in Blackboard Module*

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Complete exam debrief reflection survey on Canvas

## 6 Looking Ahead

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Quiz 9 Next Week on Canvas